SENSOR FUSION PROJECT

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1 Input/Output

Input: time, gyroZ, magX, magY, gnssHead, isNewGnss, isGoodGnss, gnssHeadAcc
Output: yawEst, gyroBias, isUseGyro, isUseMag, isUseGnss

2 Related Formulation

a) State space model

$$\begin{aligned} \dot{x} &= Ax + Bu \\ z &= Hx \end{aligned} \tag{1}$$

where:

$$A = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad x = \begin{bmatrix} \psi \\ b \end{bmatrix}, \quad u = gyroZ$$

b) State prediction

$$\begin{aligned} \hat{x}+ &= (A\hat{x} + Bu)\Delta t \\ \hat{P}+ &= (AP + PA^T + Q)\Delta t \end{aligned} \tag{2}$$

where, Q is the diagonal process noise matrix

c) Update

$$\begin{aligned} S &= HPH^T + R \\ K &= PH^T S^{-1} \\ Inov &= z - H\hat{x} \\ \hat{x}+ &= K * Inov \\ \hat{P}- &= KHP \end{aligned} \tag{3}$$

Updated by Magnetometer: $z = z_{mag} = -atan2(magY, magX), R = R_{mag}$
Updated by GNSS: $z = z_{GNSS} = gnssHead, R = gnssHeadAcc^2$

d) Angle processing function

$wrapToPi$: Limit the heading angle to the range $[-\pi, \pi]$
$mod\,(\alpha * (180/\pi), 360)$: Convert heading angle from $[-180^0, 180^0]$ to $[0^0, 360^0]$ range

e) Low pass filter

$y_k = \alpha * x_k + (1 - \alpha) * y_k - 1$

f) Abnormal rejection

$P+ = \dfrac{P + P^T}{2}$: Drive P matrix to be symmetric positive-definite
$\dfrac{|Inov|^2}{S} \geq threshold$: Inovation gating

3 Functions in Class

a) YawNav()

Initializes the internal state vector, covariance matrices, and operational thresholds of the Kalman filter.

b) pushNewData(time, gyroZ, magX, magY, gnssHead, isNewGnss, isGoodGnss, gnssHeadAcc)

Functions as the primary execution orchestrator of the sensor fusion pipeline at each time step.

c) getEst()

Provides a secure read-out interface to extract the finalized estimation states and sensor health diagnostics.

d) predictKalman(gyroZ)

Executes the Kalman Filter Prediction equations to project the state forward in time.

e) updateKalman(z, R)

Executes the Kalman Filter Measurement Correction equations to adjust the state using absolute reference data.

f) preprocessTime(currentTime)

Calculates the precise sampling interval Δt and monitors system timing integrity.

g) preprocessGyro(gyroZ)

Diagnoses the hardware health of the gyroscope to detect data freezing or connection loss.

h) preprocessMagnet(magX, magY)

Filters high-frequency magnetic noise and checks for localized magnetic distortions.

i) preprocessGnss(isNewGnss, isGoodGnss, gnssHeadAcc)

Validates, conditions, and dynamically weighs incoming GNSS heading data before triggering a measurement update.

4 Flow Chart

a) Overall flowchart

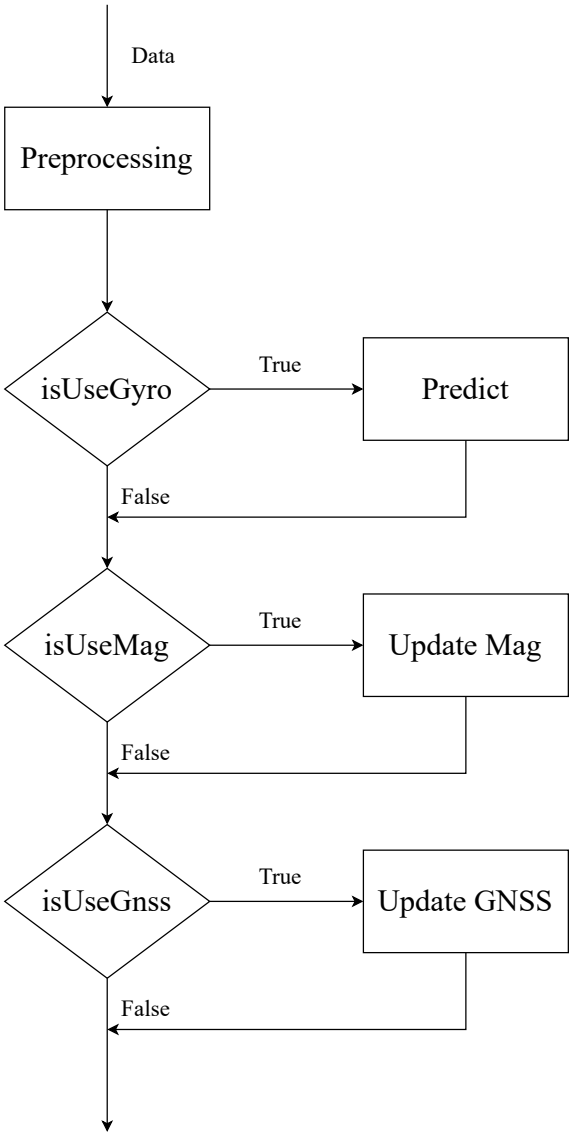


Figure 1: *Overall flowchart*

b) pushNewData

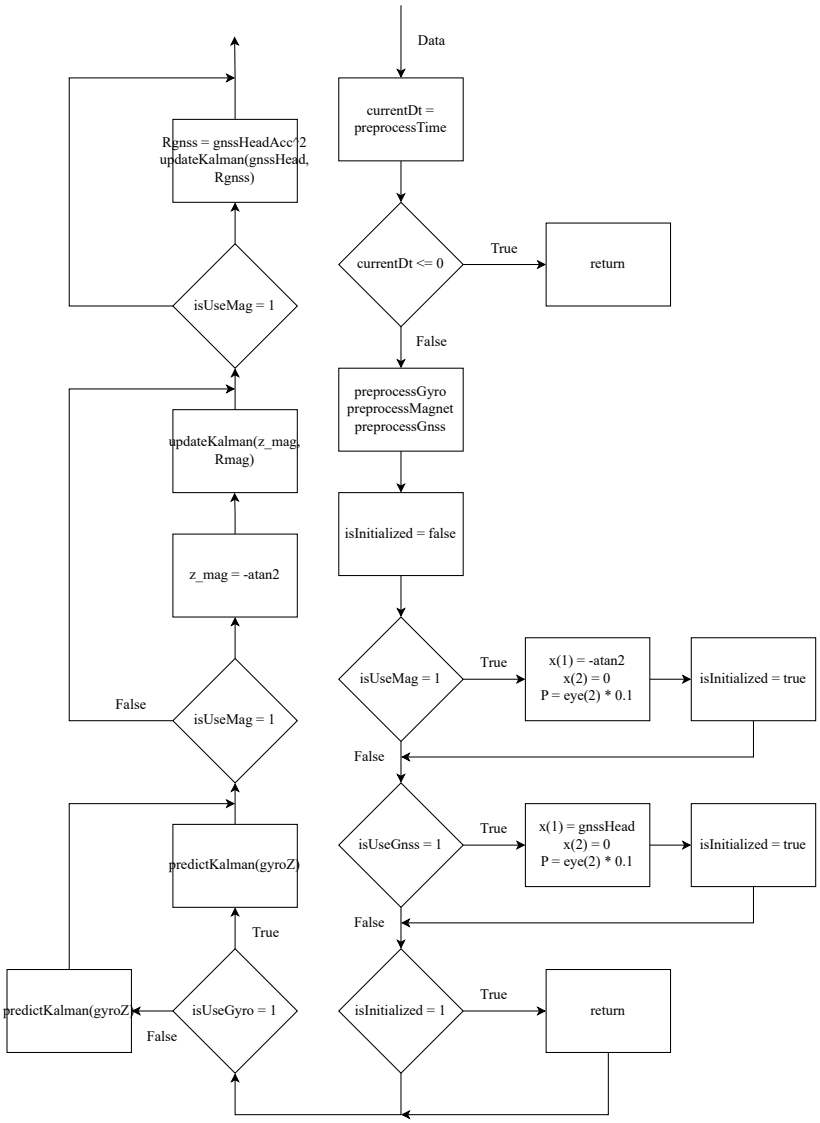


Figure 2: *pushNewData*

c) predictKalman

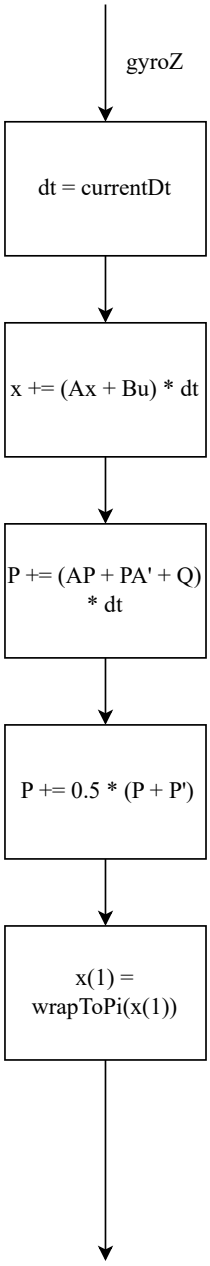


Figure 3: *predictKalman*

d) updateKalman

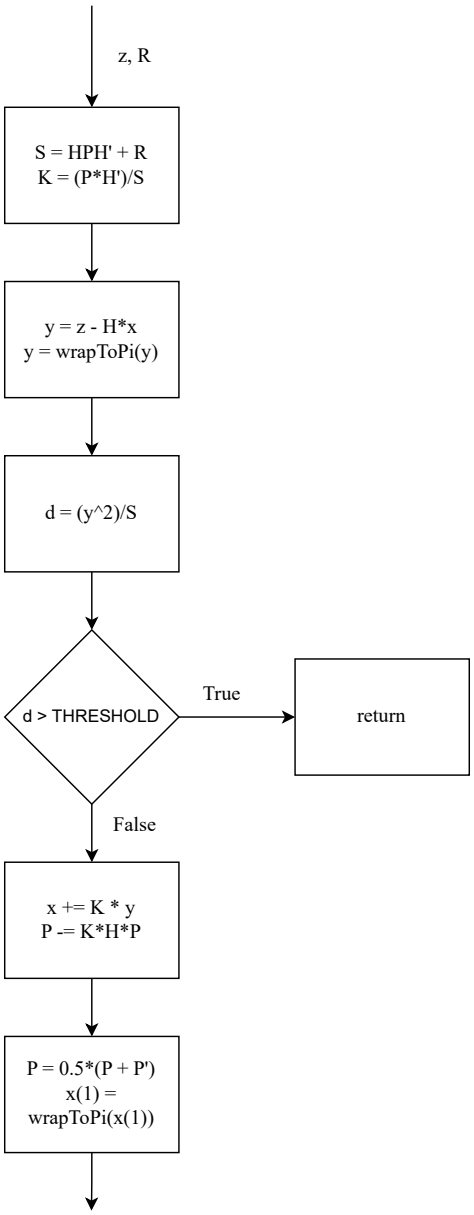


Figure 4: *updateKalman*

e) preprocessTime

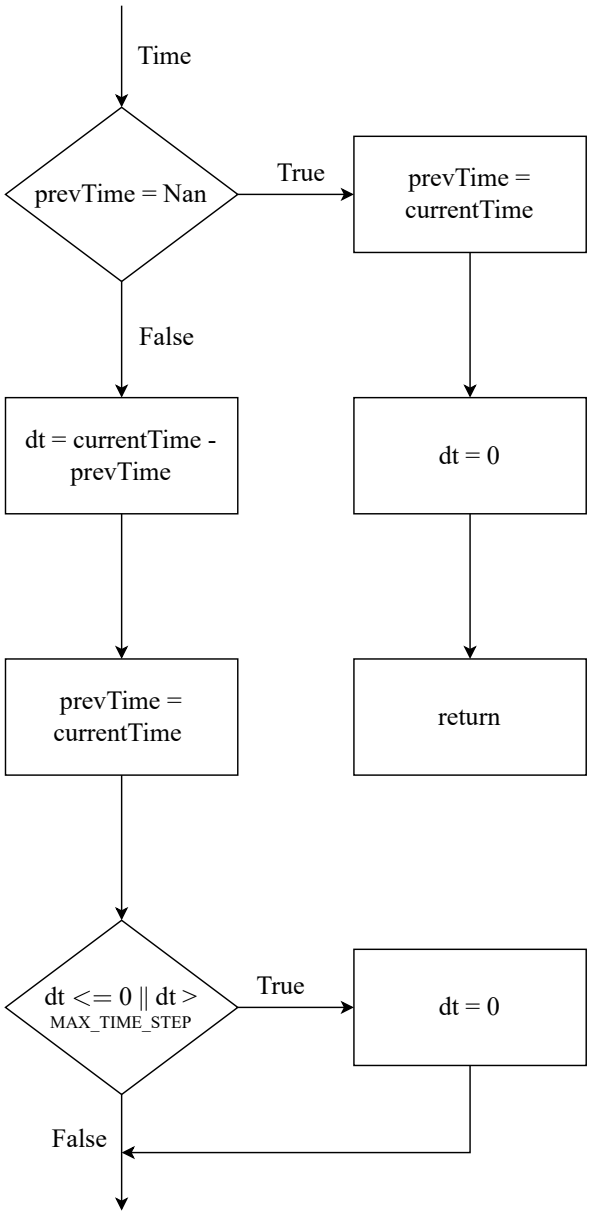


Figure 5: *preprocessTime*

f) preprocessGyro

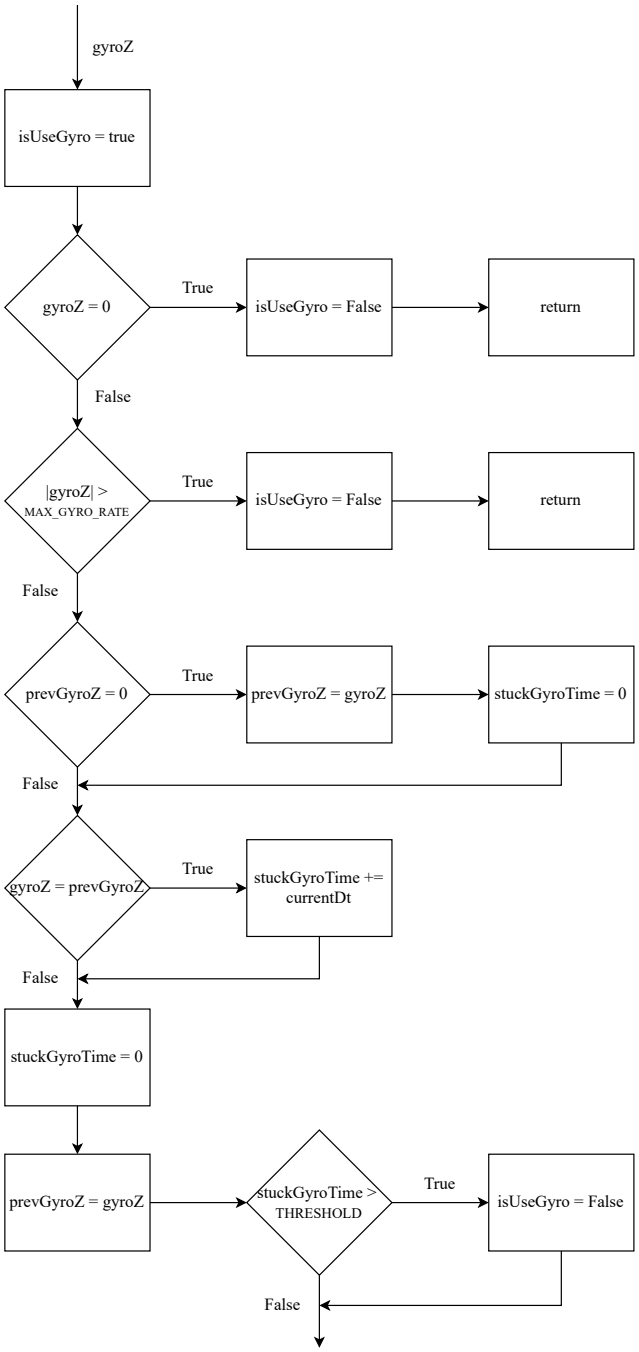


Figure 6: *preprocessGyro*

g) preprocessMagnet

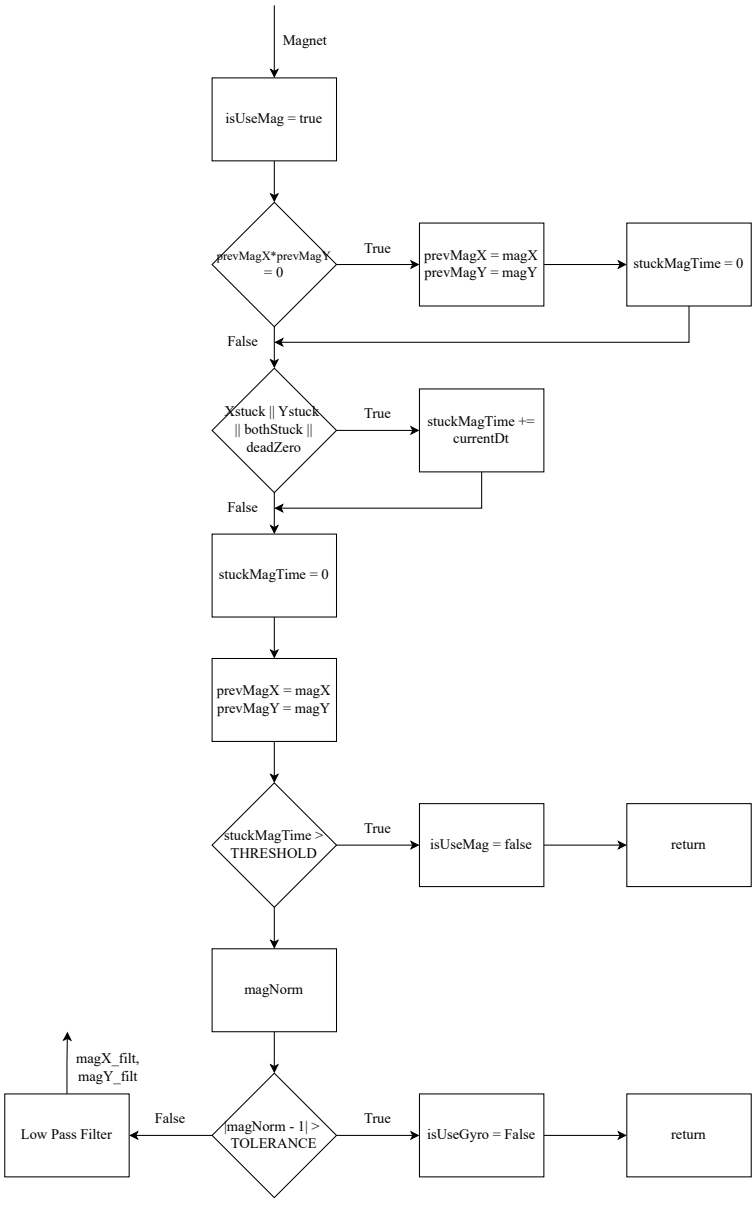


Figure 7: *preprocessMagnet*

h) preprocessGnss

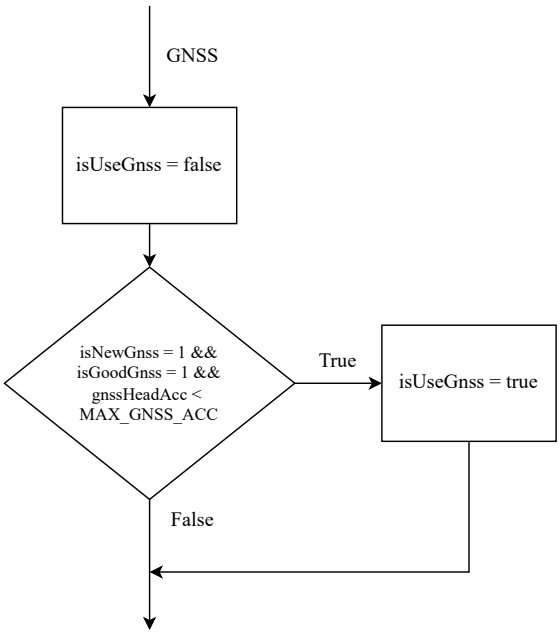


Figure 8: *preprocessGnss*

5 Results

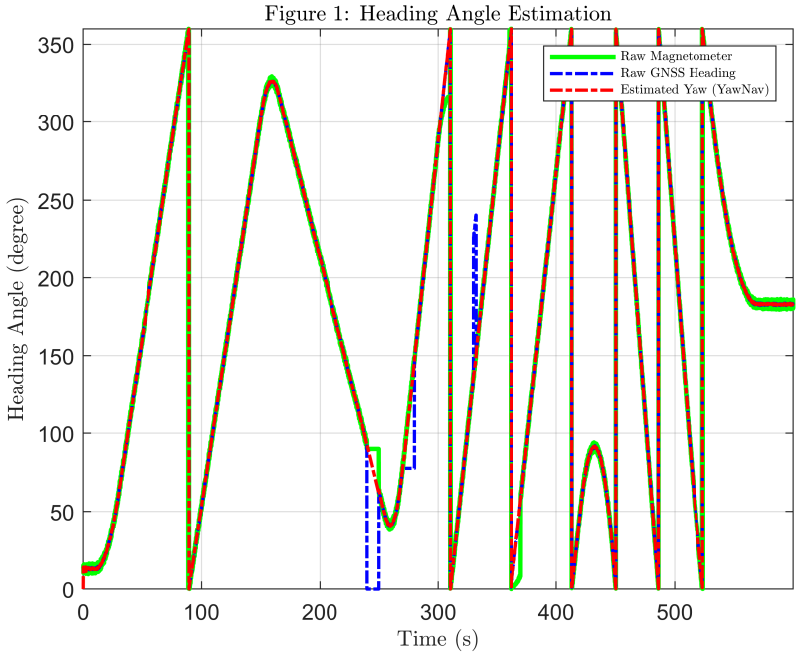


Figure 9: *Comparison of estimated heading angle and raw data*

Figure 2: Gyro Bias axis Z

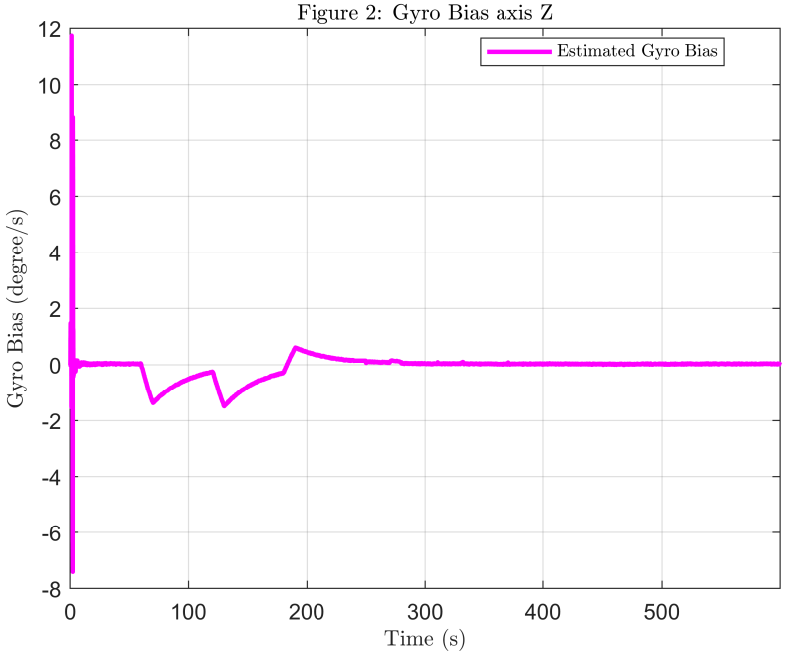


Figure 10: *Estimated gyro bias*

Figure 3: Health Flag

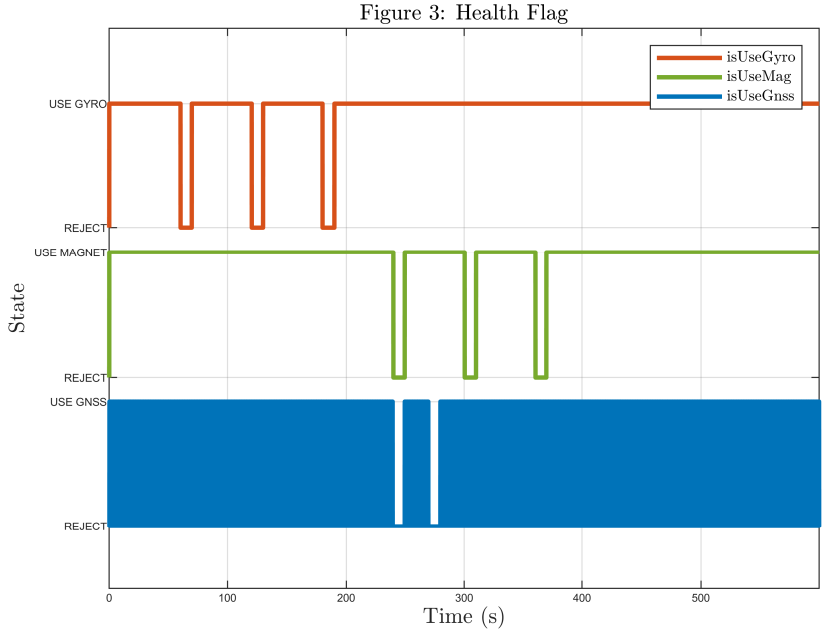


Figure 11: *Health flags*

6 Evaluation

High-Fidelity Tracking: The YawNav filter tracks the true heading with exceptional accuracy. Output normalization within the 0° to 360° range ensures smooth boundary wrap-around transitions without discontinuities.

Online Bias Calibration: The algorithm actively estimates and compensates for the dynamic gyroscope drift bias (b) in real-time. This completely eliminates open-loop integration drift and maintains long-term trajectory integrity.

Robust Preprocessing: The preprocessing stage successfully safeguards the filter against severe hardware faults, including frozen data (stuck sensors), telemetry dropouts (Δt watchdog), and low-accuracy satellite signals.

Chi-Square Gating and Outlier Rejection: Through real-time Innovation Testing, the filter rejects severe sensor spikes (e.g., GNSS anomaly at 330s) and isolates localized magnetic distortions by safely coasting on pure inertial dead-reckoning.

Numerical Stability: Enforcing symmetry and positive definiteness on the error covariance matrix via